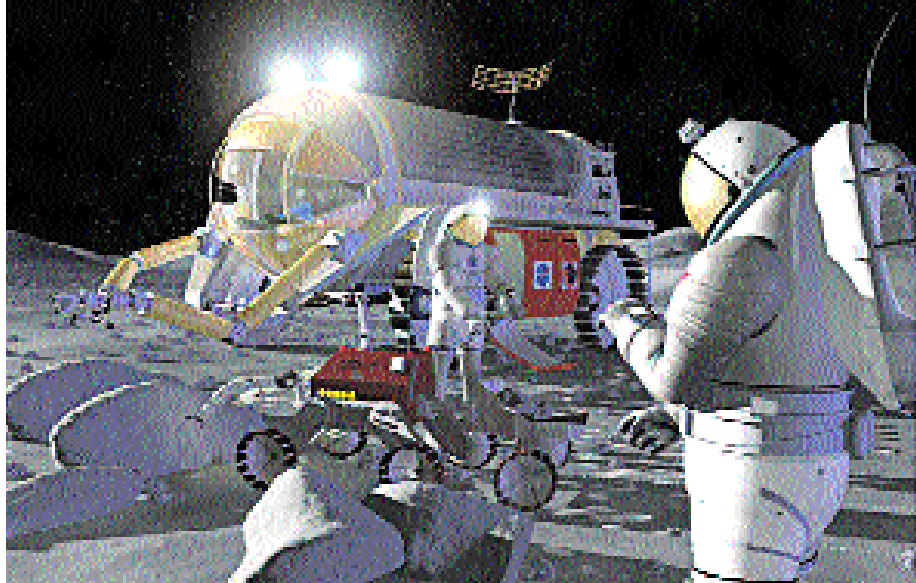




Liquid Metal?

NASA, the California Institute of Technology and the US Government invented a metal that had the properties of plastic. This will be used to build various components of space shuttles. Not only was this new form of metal more elastic, it was also much stronger than the highest grade titanium alloys. This is how Liquidmetal Inc. (www.liquidmetal.com) was born.

Unlike normal metals, Liquidmetal doesn't have a crystalline atomic structure; instead it has an amorphous atomic structure similar to plastics. This gives it high strength, hardness, elasticity and corrosion resistance. In addition, Liquidmetal has an amazing strength-to-weight ratio, making it extremely light, yet stronger than titanium.

Apart from building better hulls or components, Liquidmetal has found its way into sports goods such as racquets (www.head.com), baseball bats (www.rawlings.com) and even a TAG Heuer (www.tagheuer.com) watch series! The watch casing is made of Liquidmetal and is scratchproof, dent-resistant and doesn't corrode, while the sports equipment offers enhanced power because of Liquidmetal's elasticity and strength advantage.

Another well-known manufacturer that was drawn to Liquidmetal was Vertu. Their latest Ascent phone collection has a Liquidmetal bezel and battery cover.

Imaging

One important aspect of space exploration is imaging. Since we send robots and computer-driven ships to the farthest reaches of the solar system and beyond, which then transmit data (generally images and video) back to us, we have to make sure the imaging technology is top-notch.

Now, lenses and cameras have their own limitations and extracting still images from video always results in blurry images—since video uses frames to deliver smooth-looking movement, most video frames depict movement as blurs. But what if we could use software instead of hardware to calculate the movement and extract crystal-clear stills from video?

This not only applies to space then, and could also be used to extract great stills from your home videos. That's exactly what RedHawk Vision's Paparazzi software does!

The imaging algorithm that extracts still images from videos was developed by RedHawk (www.redhawkvision.com) and NASA's Marshall Space Flight Center. The algorithm was called the 'Real-Time Self-Contained Image-Motion Compensation for Spaceborne Imaging Instruments' algorithm—don't blame us, we didn't name it!

Basically, the algorithm calculates the movement in a scene, and repositions similar images to form a clear still picture of movement. At times it can create very psychedelic looking stills, as you can see in the examples provided. All you have to do is buy a Paparazzi Flash drive unit, connect it to a USB port, double-click the Paparazzi icon, and then play videos in Apple's QuickTime media player. Paparazzi automatically analyses the video and generates stills!

Using Paparazzi, you can now extract that favourite pose from a home video, and actually get decent enough quality to frame it and hang it on a wall! Meanwhile, NASA uses the technology to extract distortion-free images from its probes traveling through the galaxy—even extracting images from completely garbled video transmissions!

